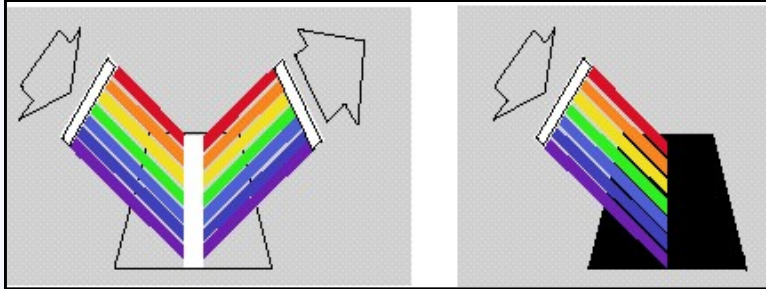


TECHNICAL DATA SHEET (FOR EXPERTS / EDUCATORS) – WORKSHOP 1 : OPTRONICS & THERMODYNAMICS OF MATERIALS

Light Reflection and Thermal Absorption



Experiment with lasers aimed at balloons



Background : this educational workshop demonstrates the interaction between highly concentrated light (laser) and polymeric material (latex) using balloons subjected to mechanical stress.

Objectives of the experiment : to visually demonstrate the difference between the reflection and thermal absorption of light depending on the balloon's color, and to illustrate spectral selectivity using the "double balloon" experiment.

I. Physical Phenomena in Focus

- Spectral absorptivity (A_λ) and thermal heating : when laser radiation of a specific wavelength λ (e.g., a green laser at 532 nm or a blue laser at 450 nm) strikes the surface of a balloon, a portion of the incident optical power P_{inc} is absorbed and converted into thermal energy. The absorbed heat flux is governed by the law of conservation of energy : $P_{abs}(\lambda) = A(\lambda) \cdot P_{inc}$
- Black rubber has an absorptivity coefficient $A \approx 1$ (behavior similar to that of a blackbody), causing an instantaneous local temperature rise up to the polymer's thermal melting point. White rubber reflects nearly all of the flux ($A \approx 0$).
- Mechanical rupture due to tensile stress : The latex of the inflated balloon is subjected to high bidirectional surface and mechanical tension. The local temperature rise induced by the laser beam weakens the elastic molecular bonds (polymers). The residual peripheral tensile stress causes an instantaneous and explosive tear in the membrane.
- Direct optical transmission (spectral selection) : In the "double balloon" experiment, the outer membrane of the white balloon is transparent to the laser's wavelength. The photons pass through the white envelope without substantial thermal interaction (high reflectance **R** and high transmittance **T**) and focus their energy solely on the inner black membrane, which absorbs all the energy and explodes.

II. Experimental Protocol and Technical Alignment

1. Spatial Focusing : Use stable tripods to eliminate operator shake and focus the laser's power density (W/m^2) on an extremely precise point on the latex.
2. Geometric nesting (double balloon) : Insert the uninflated black balloon into the white balloon using a smooth rod. Inflate the inner black balloon first, tie off its base, then inflate the outer white balloon to create a protective, transparent air chamber.

DID YOU KNOW THAT ?

Did you know that ? The secret of light that cuts through matter ! Why does a green laser destroy a black balloon in a flash, but leave a white balloon completely intact ? It's all a matter of color and energy storage ! Black is a veritable "vacuum cleaner" for light : it absorbs all the laser's photons and instantly transforms them into scorching heat that melts the stretched plastic. White, on the other hand, acts like a perfect mirror and reflects the light without heating up. By slipping a black balloon inside a transparent white balloon, you're performing a surgical experiment : the laser passes through the white balloon without touching it and causes the black balloon inside to explode ! It is precisely this thermal principle that allows engineers to use industrial lasers to cut steel with microscopic precision and surgeons to operate on the human eye safely (Academic source: IEEE Journal of Quantum Electronics).